MEHMET HAKLIDIR, Ph.D., SMIEEE

AI Executive | Founding Director, National AI Institute | AI Governance & Strategy Leader
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PROFESSIONAL SUMMARY

AI executive and national technology strategist with 24+ years of experience across artificial intelligence, autonomous systems, big data, simulation, and technology policy. Founding Director of Türkiye's first national AI institute, leading six competence centers and a 116-organization AI ecosystem. Contributed to Türkiye's first National AI Strategy and led the technical preparation and coordination of the National AI Action Plan 2026-2030 through broad ecosystem consultation. Active in international AI governance through OECD, GPAI, UNESCO, NATO, and European AI initiatives. Pioneered Türkiye's first 9-billion-parameter LLM; current research focuses on LLMs, AI safety, agentic systems, deep reinforcement learning, robotics, and autonomous decision-making under uncertainty. 55+ peer-reviewed publications.

CORE COMPETENCIES

AI Strategy & Policy Development | National AI Governance | Large Language Models (LLM) | AI Safety & Trustworthy AI | Agentic AI Systems | Deep Reinforcement Learning | Robotics & Autonomous Systems | R&D Portfolio Management | EU Funding (Horizon Europe, Digital Europe, ECSEL) | Public-Private Partnerships | Ecosystem Building | Team Leadership (130+ FTE) | Budget Management (EUR 10M+) | International Stakeholder Relations | AI Ethics & Regulatory Compliance (EU AI Act) | Big Data Architecture | Digital Twin Systems | Federated Learning | Defense Modeling & Simulation

PROFESSIONAL EXPERIENCE

Founding Director, Artificial Intelligence Institute

TÜBİTAK BİLGEM (Türkiye's national research center) | May 2023 - Present

- Founded and direct Türkiye's first national AI institute with 6 competence centers: FinTech, Intelligent Manufacturing, Smart Agriculture & Food, Climate/Sustainability, E-Commerce Technologies, and Smart Education.
- Built and mobilized a national AI ecosystem of 116 partner organizations (39 industry end-users, 42 SMEs, 26 universities, 9 public agencies); launched 50+ co-development projects mapped to UN SDGs.
- Led the technical preparation and coordination of Türkiye's National AI Action Plan 2026-2030, finalized through national ecosystem consultation.
- Pioneered BILGE, Türkiye's first 9-billion-parameter large language model; currently coordinating the multi-national Turkic LLM initiative under the Organization of Turkic States.
- Designed and oversee Türkiye's Trustworthy-AI Algorithmic Accountability Framework and 'Reliable AI' certification program for public-sector AI deployments.
- Secured EUR 2M+ in Horizon Europe and Digital Europe funding; serve as Task Leader for AI Act Compliance & Ethical AI in the BSC AI Factory project (EuroHPC JU).
- Received Gold Ecosystem Contribution Award (AI Factory Awards 2024).

Deputy Director

TÜBİTAK BİLGEM BTE | April 2022 - February 2025

- Directed 3 departments (Cloud Computing & Big Data, Data Valorization, Software Engineering) with 130 researchers and engineers and a EUR 10M portfolio budget.
- Launched Türkiye's largest big-data platform for the Customs Enforcement Directorate: 80 TB/week ingestion with 98% ML-based risk detection accuracy; won 'Best in Future of Intelligence' (IDC Future Enterprise Awards 2022) and 'Best in Machine Learning' (Future of Cloud & AI Summit 2023).
- Delivered the world's first autonomous 5G border crossing in partnership with Ford Otosan and Turkcell (EU H2020 5G-MOBIX).

- Co-developed the world's first EEG-controlled AI exoskeleton with the German Research Center for AI (DFKI).
- Deployed DerinGÖRÜ, an AI-driven public-safety suite that reunited 456 missing children with their families within 3 weeks following the 2023 earthquake.
- Co-authored OECD's 'Advancing Accountability in AI' as a member of the OECD ONE AI Expert Group.

Head of Cloud Computing and Big Data Research Lab (B3LAB)
TÜBİTAK BİLGEM | September 2020 - August 2023

- Led a 40-person multidisciplinary team delivering 12 concurrent R&D and industrial projects across cloud, big data, IoT, AI, and digital twins.
- Secured EUR 4M+ in Horizon 2020 and IPA funding by authoring and winning 5 proposals in smart manufacturing and public-sector digital transformation.
- Delivered the Safir national platform suite: Safir Bulut (cloud infrastructure), Safir Büyük Veri (big data), Safir Zeka (machine learning as a service).
- Shaped the founding vision, operating model, and national ecosystem engagement framework of the TÜBİTAK BİLGEM Artificial Intelligence Institute.
- Co-authored the 'OECD Framework for the Classification of AI Systems' (OECD ONE AI Expert Group).
- Served on national advisory boards for the AI Roadmap and the Cloud & Big Data Technologies Roadmap.

Chief Researcher / Big Data & AI Team Leader
TÜBİTAK BİLGEM B3LAB | January 2018 - September 2020

- Built the lab's AI and data science capability from the ground up: team formation, technical direction, and core infrastructure planning.
- Delivered Safir Zeka, a scalable MLaaS platform; designed Lambda Architecture reference big-data pipelines for national-scale predictive analytics, classification, and anomaly detection.

Chief Researcher / Project Manager, Defense Simulation Systems
TÜBİTAK BİLGEM | May 2005 - January 2018

- Deputy Project Manager for the National Submarine Command & Control System (MÜREN Preveze); Team Lead/Product Owner for the Submarine Tactical Simulator (DATAS); Lead Systems Engineer for Torpedo Countermeasure Simulation (DAKA-Y).
- Developed Computer Generated Forces (CGF) and tactical AI behaviors for anti-submarine warfare training within VR-Forces architecture; built HLA/RTI distributed simulation environments.
- Developed dynamic models and control/autonomy algorithms for underwater platforms: guidance, navigation, collision avoidance, and motion planning.
- Received Ship Survivability Solution Award (ISS 2010).

Assistant Product Manager / R&D Engineer
Bosch Turkey | 2000 - 2003

- Marketing analytics, data analysis and reporting, project management; SAP Key User. R&D engineer on a special development project.

EDUCATION

PhD, Control and Automation Engineering
Istanbul Technical University | 2011 - 2022

Thesis: Deep Reinforcement Learning for Partially Observable Markov Decision Processes. GPA: 3.80/4.00

MSc, Mechatronics Engineering
Istanbul Technical University | 2003 - 2006

BSc, Mechanical Engineering (Minor: Computer Science)
Istanbul University | 1999 - 2003

ADVISORY AND GOVERNANCE ROLES

- Steering Committee Member, Science, Technology and Innovation Policies Council, Presidency of the Republic of Türkiye (GenAI/AGI) - 2025-Present
- Member, Ethics Supreme Board, Ministry of National Education - 2025-Present
- Member, OECD Network of AI Experts (ONE AI) - 2020-Present
- Member, OECD.AI Index Expert Group - 2024-Present
- Expert, Global Partnership on AI (GPAI) - 2023-Present
- Delegate, NATO Data and AI Review Board (DAIRB)
- Member State Representative (Turkiye), UNESCO AI Ethics
- National Contact Point of Turkey, IRCAI (UNESCO International Research Centre on AI) - 2021-Present
- Member, European AI Alliance - 2019-Present
- Task Leader (AI Act Compliance & Ethical AI), BSC AI Factory, EuroHPC JU

SELECTED ACHIEVEMENTS

- AI for the Public Sector Award (Kamu İçin Yapay Zekâ Ödülü), received as founding Director on behalf of the TÜBİTAK BİLGEM AI Institute - AIPA 2026
- Senior Member, IEEE - 2026
- Ranked #120, World's Most Influential Voices in AI - Favikon Top 200 (January 2026)
- Gold Ecosystem Contribution Award - AI Factory Awards 2024
- Best in Machine Learning - Future of Cloud & AI Summit 2023
- Best in Future of Intelligence - IDC Future Enterprise Awards 2022
- Runner-up, Turkey Cloud Technology Awards - IDC 2020
- Ship Survivability Solution Award - ISS 2010

PUBLICATIONS AND RESEARCH

55+ peer-reviewed publications in SCI-indexed journals (IEEE Access, Natural Resources Research, Environmental Earth Sciences, Journal of Intelligent Manufacturing, Ethics and Information Technology) and leading conferences (ICRA, CVPR, CHI, LREC workshops). h-index: 11; 363+ citations. Full list: haklidir.github.io/publications

Current research: large language models, AI safety with emphasis on agentic AI systems, deep reinforcement learning, robotics, and autonomous decision-making under uncertainty.

Co-author of international AI policy documents: OECD Framework for the Classification of AI Systems (2022), OECD Advancing Accountability in AI (2023), WEF Human-Centred AI for Human Resources (2021).

TECHNICAL SKILLS

AI/ML: PyTorch, TensorFlow, Keras, Deep Reinforcement Learning (SAC, PPO, DQN), LLMs, Transformers, GANs, Federated Learning

Data & Cloud: Hadoop, Spark, Kafka, Hive, MongoDB, Lambda Architecture, OpenStack-based cloud platforms

Simulation & Robotics: HLA/RTI, VR-Forces, MATLAB/Simulink, Modelica/Dymola, Digital Twins, ROS

Programming: Python, C/C++, Java, MATLAB

Languages: Turkish (native), English (C2 - full professional), German (B2 - limited working)